

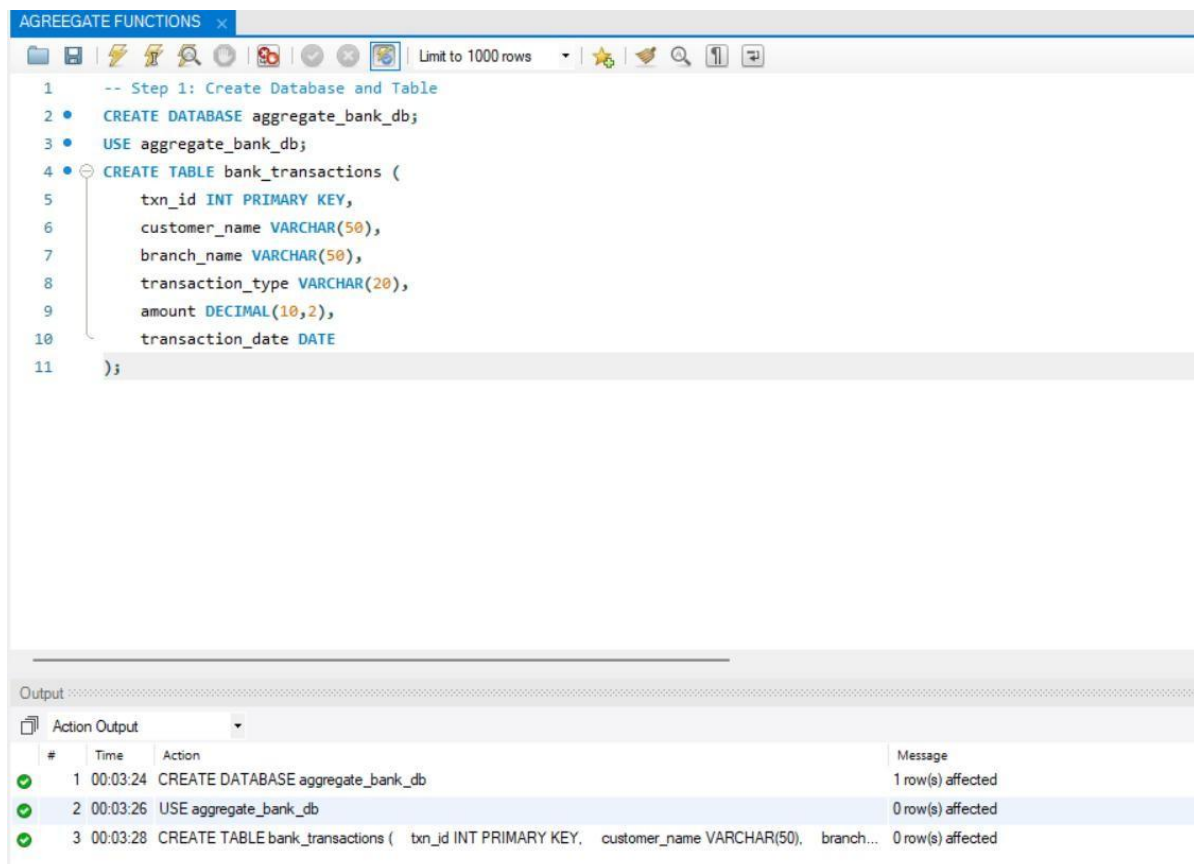
# AGREEGATE FUNCTIONS

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## Question :

Perform SQL aggregate operations on bank transaction data using SUM(), AVG(), MAX(), MIN(), and COUNT() along with WHERE, GROUP BY, HAVING, and ORDER BY clauses to analyse transaction amounts, branches, deposits, withdrawals, and transaction counts.

## Step 1 : Create Database and Table



The screenshot shows a SQL IDE window titled "AGREEGATE FUNCTIONS". The main editor displays the following SQL script:

```
-- Step 1: Create Database and Table
1  CREATE DATABASE aggregate_bank_db;
2  USE aggregate_bank_db;
3  CREATE TABLE bank_transactions (
4      txn_id INT PRIMARY KEY,
5      customer_name VARCHAR(50),
6      branch_name VARCHAR(50),
7      transaction_type VARCHAR(20),
8      amount DECIMAL(10,2),
9      transaction_date DATE
10 );
11
```

Below the editor, the "Output" tab is selected, showing the "Action Output" table:

| #   | Time     | Action                                                                                        | Message           |
|-----|----------|-----------------------------------------------------------------------------------------------|-------------------|
| ✓ 1 | 00:03:24 | CREATE DATABASE aggregate_bank_db                                                             | 1 row(s) affected |
| ✓ 2 | 00:03:26 | USE aggregate_bank_db                                                                         | 0 row(s) affected |
| ✓ 3 | 00:03:28 | CREATE TABLE bank_transactions ( txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch... | 0 row(s) affected |

JoinsSQL SQL File 3\*

```

10 transaction_date DATE
11 );
12 -- Step 2: Insert Data
13 • INSERT INTO bank_transactions VALUES
14 (101,'Angela','Hyderabad','Deposit',5000,'2024-01-05'),
15 (102,'Smija','Hyderabad','Withdrawal',2000,'2024-01-06'),
16 (103,'Joel','Vijayawada','Deposit',12000,'2024-01-08'),
17 (104,'Tianne','Vizag','Deposit',8000,'2024-01-10'),
18 (105,'Riya','Hyderabad','Withdrawal',3500,'2024-01-11'),
19 (106,'Andrea','Vizag','Deposit',15000,'2024-01-12'),
20 (107,'Advika','Vijayawada','Withdrawal',1000,'2024-01-13'),
21 (108,'Priya','Hyderabad','Deposit',9000,'2024-01-14'),
22 (109,'Nikshitha','Vizag','Withdrawal',4000,'2024-01-15'),
23 (110,'Keerthi','Vijayawada','Deposit',11000,'2024-01-16');
24 -- Step 3: SUM()
25 • SELECT SUM(amount) AS Total_Amount
26 FROM bank_transactions;
27 -- Step 4: AVG()
28 • SELECT AVG(amount) AS Average_Transaction
29 FROM bank_transactions;
30 -- Step 5: MAX()

```

Output

| # | Time     | Action                                                                                                          | Message                                                  |
|---|----------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1 | 22:02:48 | CREATE DATABASE aggregate_bank_db                                                                               | 1 row(s) affected                                        |
| 2 | 22:02:51 | USE aggregate_bank_db                                                                                           | 0 row(s) affected                                        |
| 3 | 22:02:55 | CREATE TABLE bank_transactions ( txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...                   | 0 row(s) affected                                        |
| 4 | 22:03:01 | INSERT INTO bank_transactions VALUES (101,'Angela','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Smija','H... | 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 |

## Step 2 : AGREEGATE FUNCTIONS

JoinsSQL SQL File 3\*

```

19 (106,'Andrea','Vizag','Deposit',15000,'2024-01-12'),
20 (107,'Advika','Vijayawada','Withdrawal',1000,'2024-01-13'),
21 (108,'Priya','Hyderabad','Deposit',9000,'2024-01-14'),
22 (109,'Nikshitha','Vizag','Withdrawal',4000,'2024-01-15'),
23 (110,'Keerthi','Vijayawada','Deposit',11000,'2024-01-16');
24 -- Step 3: SUM()
25 • SELECT SUM(amount) AS Total_Amount
26 FROM bank_transactions;
27 -- Step 4: AVG()

```

Result Grid

| Total_Amount |
|--------------|
| 70500.00     |

Result 1 x

Output

| # | Time     | Action                                                                                                         | Message                                                  |
|---|----------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1 | 22:02:48 | CREATE DATABASE aggregate_bank_db                                                                              | 1 row(s) affected                                        |
| 2 | 22:02:51 | USE aggregate_bank_db                                                                                          | 0 row(s) affected                                        |
| 3 | 22:02:55 | CREATE TABLE bank_transactions ( txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...                   | 0 row(s) affected                                        |
| 4 | 22:03:01 | INSERT INTO bank_transactions VALUES (101,'Angela','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Smija','... | 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 |
| 5 | 22:03:29 | SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000                                        | 1 row(s) returned                                        |

# AGREGATE FUNCTIONS

```

21  (108,'Bhanu Priya','Hyderabad','Deposit',9000,'2024-01-14'),
22  (109,'Nikshitha','Vizag','Withdrawal',4000,'2024-01-15'),
23  (110,'Keerthi','Vijayawada','Deposit',11000,'2024-01-16');
24  -- Step 3: SUM()
25  • SELECT SUM(amount) AS Total_Amount
26    FROM bank_transactions;
27  -- Step 4: AVG()
28  • SELECT AVG(amount) AS Average_Transaction
29    FROM bank_transactions;

```

| Result Grid         | Filter Rows: | Export: | Wrap Cell Content: |
|---------------------|--------------|---------|--------------------|
| Average_Transaction |              |         |                    |
| 7050.000000         |              |         |                    |

## Result 2

| # | Time     | Action                                                                                                         | Message                                                  |
|---|----------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1 | 00:03:24 | CREATE DATABASE aggregate_bank_db                                                                              | 1 row(s) affected                                        |
| 2 | 00:03:26 | USE aggregate_bank_db                                                                                          | 0 row(s) affected                                        |
| 3 | 00:03:28 | CREATE TABLE bank_transactions ( txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...                   | 0 row(s) affected                                        |
| 4 | 00:05:23 | INSERT INTO bank_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Chinni... | 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 |
| 5 | 00:06:32 | SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000                                        | 1 row(s) returned                                        |
| 6 | 00:07:11 | SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000                                 | 1 row(s) returned                                        |

# AGREGATE FUNCTIONS

```

24  -- Step 3: SUM()
25  • SELECT SUM(amount) AS Total_Amount
26    FROM bank_transactions;
27  -- Step 4: AVG()
28  • SELECT AVG(amount) AS Average_Transaction
29    FROM bank_transactions;
30  -- Step 5: MAX()
31  • SELECT MAX(amount) AS Highest_Transaction
32    FROM bank_transactions;

```

| Result Grid         | Filter Rows: | Export: | Wrap Cell Content: |
|---------------------|--------------|---------|--------------------|
| Highest_Transaction |              |         |                    |
| 15000.00            |              |         |                    |

## Result 3

| # | Time     | Action                                                                                                      | Message                                                  |
|---|----------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 2 | 00:03:26 | USE aggregate_bank_db                                                                                       | 0 row(s) affected                                        |
| 3 | 00:03:28 | CREATE TABLE bank_transactions ( txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...                 | 0 row(s) affected                                        |
| 4 | 00:05:23 | INSERT INTO bank_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Chi... | 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 |
| 5 | 00:06:32 | SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000                                     | 1 row(s) returned                                        |
| 6 | 00:07:11 | SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000                              | 1 row(s) returned                                        |
| 7 | 00:07:47 | SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000                              | 1 row(s) returned                                        |



## Step 3 : WHERE + AGGREGATE

AGGREGATE FUNCTIONS

```
34 • SELECT MIN(amount) AS Lowest_Transaction
35 FROM bank_transactions;
36 -- Step 7: COUNT()
37 • SELECT COUNT(*) AS Total_Transactions
38 FROM bank_transactions;
39 -- Step 8: WHERE + SUM()
40 • SELECT SUM(amount) AS Total_Deposit
41 FROM bank_transactions
42 WHERE transaction_type = 'Deposit';
```

Result Grid

| Total_Deposit |
|---------------|
| 60000.00      |

Result 6

Output

Action Output

| #  | Time     | Action                                                                                               | Message           |
|----|----------|------------------------------------------------------------------------------------------------------|-------------------|
| 5  | 00:06:32 | SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000                              | 1 row(s) returned |
| 6  | 00:07:11 | SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000                       | 1 row(s) returned |
| 7  | 00:07:47 | SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000                       | 1 row(s) returned |
| 8  | 00:08:17 | SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000                        | 1 row(s) returned |
| 9  | 00:08:53 | SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000                           | 1 row(s) returned |
| 10 | 00:09:40 | SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIM... | 1 row(s) returned |

## Step 4 : GROUP BY

AGGREGATE FUNCTIONS

```
39 -- Step 8: WHERE + SUM()
40 • SELECT SUM(amount) AS Total_Deposit
41 FROM bank_transactions
42 WHERE transaction_type = 'Deposit';
43 -- Step 9: GROUP BY + SUM()
44 • SELECT branch_name,
45 SUM(amount) AS Total_Amount
46 FROM bank_transactions
47 GROUP BY branch_name;
```

Result Grid

| branch_name | Total_Amount |
|-------------|--------------|
| Hyderabad   | 19500.00     |
| Vijayawada  | 24000.00     |
| Vizag       | 27000.00     |

Result 7

Output

Action Output

| #  | Time     | Action                                                                                               | Message           |
|----|----------|------------------------------------------------------------------------------------------------------|-------------------|
| 6  | 00:07:11 | SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000                       | 1 row(s) returned |
| 7  | 00:07:47 | SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000                       | 1 row(s) returned |
| 8  | 00:08:17 | SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000                        | 1 row(s) returned |
| 9  | 00:08:53 | SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000                           | 1 row(s) returned |
| 10 | 00:09:40 | SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIM... | 1 row(s) returned |
| 11 | 00:10:56 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...          | 3 row(s) returned |



## Step 5 : GROUP BY + HAVING

AGREEGATE FUNCTIONS

```
45      SUM(amount) AS Total_Amount
46  FROM bank_transactions
47  GROUP BY branch_name;
48  -- Step 10: GROUP BY + HAVING
49  • SELECT branch_name,
50      SUM(amount) AS Total_Amount
51  FROM bank_transactions
52  GROUP BY branch_name
53  HAVING SUM(amount) > 20000;
```

Result Grid

| branch_name | Total_Amount |
|-------------|--------------|
| Vijayawada  | 24000.00     |
| Vizag       | 27000.00     |

Result 8

Output

Action Output

| #  | Time     | Action                                                                                                                 | Message           |
|----|----------|------------------------------------------------------------------------------------------------------------------------|-------------------|
| 7  | 00:07:47 | SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000                                         | 1 row(s) returned |
| 8  | 00:08:17 | SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000                                          | 1 row(s) returned |
| 9  | 00:08:53 | SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000                                             | 1 row(s) returned |
| 10 | 00:09:40 | SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000            | 1 row(s) returned |
| 11 | 00:10:56 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name                            | 3 row(s) returned |
| 12 | 00:12:23 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name HAVING SUM(amount) > 20000 | 2 row(s) returned |

## Step 6 : GROUP BY + ORDER BY

AGREEGATE FUNCTIONS

```
51  FROM bank_transactions
52  GROUP BY branch_name
53  HAVING SUM(amount) > 20000;
54  -- Step 11: GROUP BY + ORDER BY
55  • SELECT branch_name,
56      SUM(amount) AS Total_Amount
57  FROM bank_transactions
58  GROUP BY branch_name
59  ORDER BY Total_Amount DESC;
```

Result Grid

| branch_name | Total_Amount |
|-------------|--------------|
| Vizag       | 27000.00     |
| Vijayawada  | 24000.00     |
| Hyderabad   | 19500.00     |

Result 9

Output

Action Output

| #  | Time     | Action                                                                                                                                            | Message           |
|----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 8  | 00:08:17 | SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000                                                                     | 1 row(s) returned |
| 9  | 00:08:53 | SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000                                                                        | 1 row(s) returned |
| 10 | 00:09:40 | SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000                                       | 1 row(s) returned |
| 11 | 00:10:56 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name                                                       | 3 row(s) returned |
| 12 | 00:12:23 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name HAVING SUM(amount) > 20000                            | 2 row(s) returned |
| 13 | 00:13:21 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name HAVING SUM(amount) > 20000 ORDER BY Total_Amount DESC | 3 row(s) returned |

## Step 7: GROUP BY + HAVING + ORDER BY

AGREEGATE FUNCTIONS

```
58 GROUP BY branch_name
59 ORDER BY Total_Amount DESC;
60 -- Step 12: GROUP BY + HAVING + ORDER BY
61 • SELECT branch_name,
62       COUNT(*) AS Total_Transactions
63 FROM bank_transactions
64 GROUP BY branch_name
65 HAVING COUNT(*) >= 3
66 ORDER BY Total_Transactions DESC;
```

Result Grid

| branch_name | Total_Transactions |
|-------------|--------------------|
| Hyderabad   | 4                  |
| Vijayawada  | 3                  |
| Vizag       | 3                  |

Result 10

Output

Action Output

| #  | Time     | Action                                                                                                  | Message           |
|----|----------|---------------------------------------------------------------------------------------------------------|-------------------|
| 9  | 00:08:53 | SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000                              | 1 row(s) returned |
| 10 | 00:09:40 | SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT ... | 1 row(s) returned |
| 11 | 00:10:56 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...             | 3 row(s) returned |
| 12 | 00:12:23 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...             | 2 row(s) returned |
| 13 | 00:13:21 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...             | 3 row(s) returned |
| 14 | 00:14:11 | SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...           | 3 row(s) returned |

## Step 8 : WHERE + COUNT()

AGREEGATE FUNCTIONS

```
62 COUNT(*) AS Total_Transactions
63 FROM bank_transactions
64 GROUP BY branch_name
65 HAVING COUNT(*) >= 3
66 ORDER BY Total_Transactions DESC;
67 -- Step 13: WHERE + COUNT()
68 • SELECT COUNT(*) AS Withdrawals
69 FROM bank_transactions
70 WHERE transaction_type = 'Withdrawal';
```

Result Grid

| Withdrawals |
|-------------|
| 4           |

Result 11

Output

Action Output

| #  | Time     | Action                                                                                                | Message           |
|----|----------|-------------------------------------------------------------------------------------------------------|-------------------|
| 10 | 00:09:40 | SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIM...  | 1 row(s) returned |
| 11 | 00:10:56 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...           | 3 row(s) returned |
| 12 | 00:12:23 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...           | 2 row(s) returned |
| 13 | 00:13:21 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...           | 3 row(s) returned |
| 14 | 00:14:11 | SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...         | 3 row(s) returned |
| 15 | 00:15:22 | SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ... | 1 row(s) returned |

## Step 9 : WHERE + AVG()

AGREGATE FUNCTIONS

```
66 ORDER BY Total_Transactions DESC;
67 -- Step 13: WHERE + COUNT()
68 • SELECT COUNT(*) AS Withdrawals
69 FROM bank_transactions
70 WHERE transaction_type = 'Withdrawal';
71 -- Step 14: WHERE + AVG()
72 • SELECT AVG(amount) AS Avg_Deposit
73 FROM bank_transactions
74 WHERE transaction_type = 'Deposit';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

| Avg_Deposit  |
|--------------|
| 10000.000000 |

Result 12

Output

Action Output

| #    | Time     | Action                                                                                                | Message           |
|------|----------|-------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 11 | 00:10:56 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...           | 3 row(s) returned |
| ✓ 12 | 00:12:23 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...           | 2 row(s) returned |
| ✓ 13 | 00:13:21 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...           | 3 row(s) returned |
| ✓ 14 | 00:14:11 | SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...         | 3 row(s) returned |
| ✓ 15 | 00:15:22 | SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ... | 1 row(s) returned |
| ✓ 16 | 00:20:39 | SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT ... | 1 row(s) returned |

## Step 10 : GROUP BY + MAX()

AGREGATE FUNCTIONS

```
71 -- Step 14: WHERE + AVG()
72 • SELECT AVG(amount) AS Avg_Deposit
73 FROM bank_transactions
74 WHERE transaction_type = 'Deposit';
75 -- Step 15: GROUP BY + MAX()
76 • SELECT branch_name,
77 MAX(amount) AS Highest_Amount
78 FROM bank_transactions
79 GROUP BY branch_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

| branch_name | Highest_Amount |
|-------------|----------------|
| Hyderabad   | 9000.00        |
| Vijayawada  | 12000.00       |
| Vizag       | 15000.00       |

Result 13

Output

Action Output

| #    | Time     | Action                                                                                                | Message           |
|------|----------|-------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 12 | 00:12:23 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...           | 2 row(s) returned |
| ✓ 13 | 00:13:21 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...           | 3 row(s) returned |
| ✓ 14 | 00:14:11 | SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...         | 3 row(s) returned |
| ✓ 15 | 00:15:22 | SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ... | 1 row(s) returned |
| ✓ 16 | 00:20:39 | SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT ... | 1 row(s) returned |
| ✓ 17 | 00:21:12 | SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...           | 3 row(s) returned |



## Step 11: GROUP BY + MIN()

AGREEGATE FUNCTIONS

```
76 • SELECT branch_name,
77       MAX(amount) AS Highest_Amount
78 FROM bank_transactions
79 GROUP BY branch_name;
80 -- Step 16: GROUP BY + MIN()
81 • SELECT branch_name,
82       MIN(amount) AS Lowest_Amount
83 FROM bank_transactions
84 GROUP BY branch_name;
```

Result Grid

|   | branch_name | Lowest_Amount |
|---|-------------|---------------|
| ▶ | Hyderabad   | 2000.00       |
|   | Vijayawada  | 1000.00       |
|   | Vizag       | 4000.00       |

Result 14

Output

Action Output

| #    | Time     | Action                                                                                                | Message           |
|------|----------|-------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 13 | 00:13:21 | SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...           | 3 row(s) returned |
| ✓ 14 | 00:14:11 | SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...         | 3 row(s) returned |
| ✓ 15 | 00:15:22 | SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ... | 1 row(s) returned |
| ✓ 16 | 00:20:39 | SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...  | 1 row(s) returned |
| ✓ 17 | 00:21:12 | SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...           | 3 row(s) returned |
| ✓ 18 | 00:21:47 | SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...           | 3 row(s) returned |

## Step 12: WHERE + ORDER BY

JoinsSQL SQL File 3'

```
81 • SELECT branch_name,
82       MIN(amount) AS Lowest_Amount
83 FROM bank_transactions
84 GROUP BY branch_name;
85 -- Step 17: WHERE + ORDER BY
86 • SELECT *
87 FROM bank_transactions
88 WHERE amount > 8000
89 ORDER BY amount DESC;
```

Result Grid

|   | txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date |
|---|--------|---------------|-------------|------------------|----------|------------------|
| ▶ | 106    | Andrea        | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
|   | 103    | Joel          | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
|   | 110    | Keerthi       | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
|   | 108    | Priya         | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |
| * | NULL   | NULL          | NULL        | NULL             | NULL     | NULL             |

bank\_transactions 15

Output

Action Output

| #    | Time     | Action                                                                                                | Message           |
|------|----------|-------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 14 | 22:06:00 | SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...         | 3 row(s) returned |
| ✓ 15 | 22:06:07 | SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ... | 1 row(s) returned |
| ✓ 16 | 22:06:12 | SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...  | 1 row(s) returned |
| ✓ 17 | 22:06:18 | SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...           | 3 row(s) returned |
| ✓ 18 | 22:06:23 | SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...           | 3 row(s) returned |
| ✓ 19 | 22:06:29 | SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000                | 4 row(s) returned |

## Step 13: GROUP BY + COUNT()

AGREGATE FUNCTIONS

```
86 • SELECT *
87 FROM bank_transactions
88 WHERE amount > 8000
89 ORDER BY amount DESC;
90 -- Step 18: GROUP BY + COUNT()
91 • SELECT branch_name,
92        COUNT(*) AS Customer_Count
93 FROM bank_transactions
94 GROUP BY branch_name;
```

Result Grid

|   | branch_name | Customer_Count |
|---|-------------|----------------|
| ▶ | Hyderabad   | 4              |
|   | Vijayawada  | 3              |
|   | Vizag       | 3              |

Result 16

Output

Action Output

| # | Time        | Action                                                                                                | Message           |
|---|-------------|-------------------------------------------------------------------------------------------------------|-------------------|
| ✓ | 15 00:15:22 | SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ... | 1 row(s) returned |
| ✓ | 16 00:20:39 | SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...  | 1 row(s) returned |
| ✓ | 17 00:21:12 | SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...           | 3 row(s) returned |
| ✓ | 18 00:21:47 | SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...           | 3 row(s) returned |
| ✓ | 19 00:22:23 | SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000                | 4 row(s) returned |
| ✓ | 20 00:23:05 | SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...           | 3 row(s) returned |

## Step 14: HAVING + COUNT()

AGREGATE FUNCTIONS

```
92        COUNT(*) AS Customer_Count
93 FROM bank_transactions
94 GROUP BY branch_name;
95 -- Step 19: HAVING + COUNT()
96 • SELECT branch_name,
97        COUNT(*) AS Total_Transactions
98 FROM bank_transactions
99 GROUP BY branch_name
100 HAVING COUNT(*) > 3;
```

Result Grid

|   | branch_name | Total_Transactions |
|---|-------------|--------------------|
| ▶ | Hyderabad   | 4                  |

Result 17

Output

Action Output

| # | Time        | Action                                                                                               | Message           |
|---|-------------|------------------------------------------------------------------------------------------------------|-------------------|
| ✓ | 16 00:20:39 | SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT... | 1 row(s) returned |
| ✓ | 17 00:21:12 | SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...          | 3 row(s) returned |
| ✓ | 18 00:21:47 | SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...          | 3 row(s) returned |
| ✓ | 19 00:22:23 | SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000               | 4 row(s) returned |
| ✓ | 20 00:23:05 | SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...          | 3 row(s) returned |
| ✓ | 21 00:23:42 | SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...        | 1 row(s) returned |

## Step 15: GROUP BY + AVG()

AGREEGATE FUNCTIONS

```
97      COUNT(*) AS Total_Transactions
98  FROM bank_transactions
99  GROUP BY branch_name
100  HAVING COUNT(*) > 3;
101  -- Step 20: GROUP BY + AVG()
102  • SELECT branch_name,
103        AVG(amount) AS Avg_Amount
104  FROM bank_transactions
105  GROUP BY branch_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

|   | branch_name | Avg_Amount  |
|---|-------------|-------------|
| ▶ | Hyderabad   | 4875.000000 |
|   | Vijayawada  | 8000.000000 |
|   | Vizag       | 9000.000000 |

Result 18 x

Output

Action Output

| # | Time        | Action                                                                                        | Message           |
|---|-------------|-----------------------------------------------------------------------------------------------|-------------------|
| ✓ | 17 00:21:12 | SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...   | 3 row(s) returned |
| ✓ | 18 00:21:47 | SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...   | 3 row(s) returned |
| ✓ | 19 00:22:23 | SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000        | 4 row(s) returned |
| ✓ | 20 00:23:05 | SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...   | 3 row(s) returned |
| ✓ | 21 00:23:42 | SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_... | 1 row(s) returned |
| ✓ | 22 00:24:33 | SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...    | 3 row(s) returned |

## Step 16: WHERE + GROUP BY + HAVING

AGREEGATE FUNCTIONS

```
104  FROM bank_transactions
105  GROUP BY branch_name;
106  -- Step 21: WHERE + GROUP BY + HAVING
107  • SELECT branch_name,
108        SUM(amount) AS Total_Deposit
109  FROM bank_transactions
110  WHERE transaction_type = 'Deposit'
111  GROUP BY branch_name
112  HAVING SUM(amount) > 15000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

|   | branch_name | Total_Deposit |
|---|-------------|---------------|
| ▶ | Vijayawada  | 23000.00      |
|   | Vizag       | 23000.00      |

Result 19 x

Output

Action Output

| # | Time        | Action                                                                                         | Message           |
|---|-------------|------------------------------------------------------------------------------------------------|-------------------|
| ✓ | 18 00:21:47 | SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...    | 3 row(s) returned |
| ✓ | 19 00:22:23 | SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000         | 4 row(s) returned |
| ✓ | 20 00:23:05 | SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...    | 3 row(s) returned |
| ✓ | 21 00:23:42 | SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...  | 1 row(s) returned |
| ✓ | 22 00:24:33 | SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...     | 3 row(s) returned |
| ✓ | 23 00:25:15 | SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_t... | 2 row(s) returned |

## Step 17: GROUP BY + HAVING + ORDER BY

AGGREGATE FUNCTIONS

```
111 GROUP BY branch_name
112 HAVING SUM(amount) > 15000;
113 -- Step 22: GROUP BY + HAVING + ORDER BY
114 • SELECT branch_name,
115        AVG(amount) AS Avg_Amount
116 FROM bank_transactions
117 GROUP BY branch_name
118 HAVING AVG(amount) > 7000
119 ORDER BY Avg_Amount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

|   | branch_name | Avg_Amount  |
|---|-------------|-------------|
| ▶ | Vizag       | 9000.000000 |
|   | Vijayawada  | 8000.000000 |

Result 20

Output

Action Output

| #    | Time     | Action                                                                                         | Message           |
|------|----------|------------------------------------------------------------------------------------------------|-------------------|
| ✓ 19 | 00:22:23 | SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000         | 4 row(s) returned |
| ✓ 20 | 00:23:05 | SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...    | 3 row(s) returned |
| ✓ 21 | 00:23:42 | SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...  | 1 row(s) returned |
| ✓ 22 | 00:24:33 | SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...     | 3 row(s) returned |
| ✓ 23 | 00:25:15 | SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_t... | 2 row(s) returned |
| ✓ 24 | 00:25:51 | SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...     | 2 row(s) returned |